

MACS+ - Membrane Action Of Composite Structures In Case Of Fire

R E P O R T



1. Project properties

Project name:	Atlantic Park Phase 2 Unit 7	Prepared by:	Kevin Kurniawan
Client name:		Company name:	Fire Dynamics Group Limited
Client company:	Winvic Construction Limited	Date:	29 April 2026, 13:57
Job number:	0563	Checked by:	
Comments:	Second Floors Slab - Plant Deck		

2. General arrangement

- Spans

Span 1: 7.3 m

Span 2: 7.48 m

- Unprotected beams

Number of internal unprotected beams: 2

3. Deck details

- Deck properties

Deck: Multideck 60 Type: Trapezoidal
 Depth: 60 mm Top flange: 142 mm
 Pitch: 323 mm Bottom flange: 119 mm
 Stiffener height: 9 mm

4. Slab details

- Concrete

Concrete type: Normal Slab depth: 150 mm
 Cylinder compressive strength of concrete (f_{ck}): 30 N/mm²

- Mesh

Mesh type: A193
 Longitudinal mesh area: 193 mm²/m Bar size: 7 mm
 Transverse mesh area: 193 mm²/m Bar size: 7 mm
 Average mesh axis distance: 52 mm Mesh yield stress: 500 N/mm²

5. Beams details

- Unprotected beams

Beam type: Solid beam
 Section family: UK sections Steel grade: S355
 Degree of shear connection: 80 %
 Section size: UB 356x127x33
 Details: $h = 349$ mm, $b = 125.4$ mm, $t_w = 6$ mm, $t_f = 8.5$ mm

- Side A perimeter beam

Beam type: Solid beam
 Section family: UK sections Steel grade: S355
 Section size: UB 356x127x33
 Details: $h = 349$ mm, $b = 125.4$ mm, $t_w = 6$ mm, $t_f = 8.5$ mm
 Beam location: Internal beam Construction type: Composite
 Degree of shear connection: 80 %

- Side B perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, $t_w = 7.6$ mm, $t_f = 10.9$ mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

- Side C perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, $t_w = 6$ mm, $t_f = 8.5$ mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %

- Side D perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, $t_w = 7.6$ mm, $t_f = 10.9$ mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

6. Loading details

- Normal (Cold)

Leading variable action:	7.5 kN/m ²
Accompanying variable action:	0 kN/m ²
Dead load including beam, excluding slab:	0.5 kN/m ²
Calculated slab weight including mesh:	2.83 kN/m ²

- Fire (Hot)

Combination factor for permanent action:	1.0
Combination factor for leading variable action:	0.5
Combination factor for other variable action:	0.3

7. Fire & Analysis

- Parametric temperature-time curve

Compartment length:	37.1 m	Window height:	3.4 m
Compartment width:	7.26 m	Window length:	35.9 m
Compartment height:	3.4 m	Glazing breakage:	80.970 %
Fire load:	109.23 MJ/m ²	Wall lining (B) factor:	1700 J/m ² s ^{1/2} K
Growth rate:	Medium	Combustion factor:	0.8

8. Summary output

- Default mesh direction

Maximum unity factor:	0.23	Floor slab adequate
Factored load in fire:	7.08 kN/m ²	
Fire curve:	Parametric temperature-time curve	

9. Default mesh direction

- Tabular results

Longitudinal mesh area:	193 mm ² /m	Bar size:	7 mm
Transverse mesh area:	193 mm ² /m	Bar size:	7 mm

Factored load in fire: 7.08 kN/m²

Time	Beam	Mesh	Slab top	Slab bottom	Beam capacity	Maximum allowable deflection	Slab yield	Enhancement	Slab capacity	Total capacity	Unity factor
mins	°C	°C	°C	°C	kN/m ²	mm	kN/m ²		kN/m ²	kN/m ²	
0	20	20	20	20	27.22	158	2.13	1.75	3.73	30.94	0.23
4	22	20	20	23	27.22	159	2.13	1.76	3.74	30.95	0.23
8	27	20	20	27	27.22	160	2.13	1.76	3.75	30.97	0.23
12	36	21	20	33	27.22	162	2.13	1.77	3.77	30.98	0.23
16	46	22	20	40	27.22	164	2.13	1.78	3.79	31.00	0.23
20	57	23	20	47	27.22	166	2.13	1.79	3.81	31.02	0.23
24	50	24	20	34	27.22	162	2.13	1.77	3.77	30.98	0.23
28	43	23	21	30	27.22	161	2.13	1.77	3.75	30.97	0.23
32	38	23	21	28	27.22	160	2.13	1.76	3.75	30.96	0.23
36	33	23	21	27	27.22	160	2.13	1.76	3.74	30.96	0.23
40	30	23	21	26	27.22	159	2.13	1.76	3.74	30.96	0.23
44	28	23	21	25	27.22	159	2.13	1.76	3.74	30.95	0.23

Maximum unity factor: 0.23 **Floor slab adequate**

• Perimeter beam check

Side A	Beam type:	Solid beam	Composite Internal beam
	Section size:	UB 356x127x33	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.45	
	Critical temperature:	628 °C	
Side B	Beam type:	Solid beam	Composite Edge beam
	Section size:	UB 457x152x52	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.45	
	Critical temperature:	631 °C	
Side C	Beam type:	Solid beam	Composite Internal beam
	Section size:	UB 356x127x33	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.45	
	Critical temperature:	628 °C	
Side D	Beam type:	Solid beam	Composite Edge beam
	Section size:	UB 457x152x52	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.45	
	Critical temperature:	631 °C	

10. Graphical output

